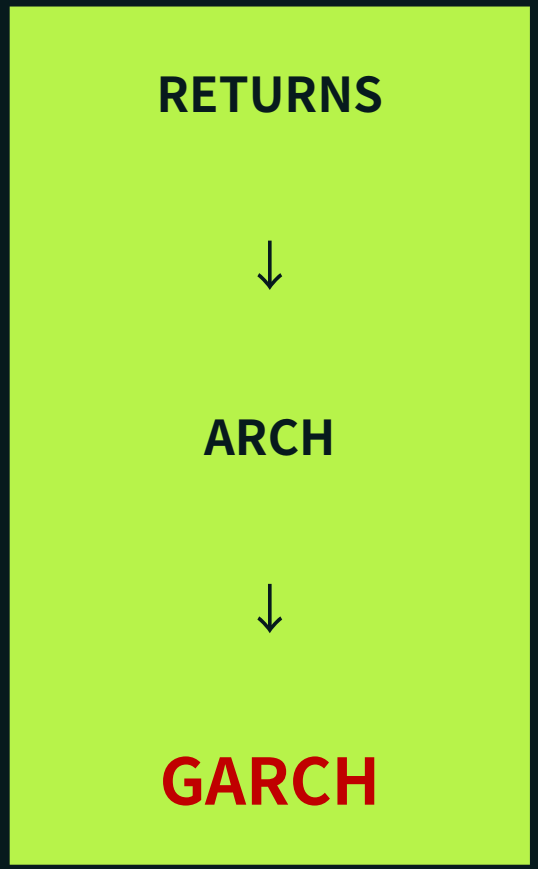
ARCH and GARCH Models
Forecasting volatility

Model the size of the swings, not the mean of returns



Advanced Financial Time Series · Week 5

From returns, we widen to ARCH, then GARCH

From models of the mean (Weeks 1–4) to models of volatility

00	Prices and returns	**What do we model?** Simple · log · continuous returns
01	Why volatility	**Volatility clusters** History · clustering · conditional variance
02	The ARCH Model	**Yesterday's shock, today's risk** ARCH(1) · one-day volatility · ARCH(q)
03	The GARCH Model	**Long memory, few parameters** GARCH(1,1) · persistence · long-run variance · forecasts
04	In practice	**Estimate and use** Extensions · maximum likelihood · VaR · volatility targeting

ARCH:GARCH = AR:ARMA — the same idea applied to variance

00

Prices and returns

Volatility models are built on returns, not prices

A return is the rate of price change, usually a log difference

p(t) = price at t; buy at t−1, sell at t (no cash dividends assumed)

Simple return ≈ log return

$$r_t = \frac{p_t - p_{t-1}}{p_{t-1}} \approx \log(p_t) - \log(p_{t-1})$$

Log returns add up to period returns

$$\log \frac{p_T}{p_0} = r_1 + r_2 + \cdots + r_T \qquad \text{(log returns add up over time)}$$

How to read it

- For small returns the two are nearly equal
- Log returns add across periods
- E.g. 100 → 101: 1% ≈ 0.995%
- From here, r(t) is the log return (%)

Prices wander (non-stationary); returns hover near 0 (stationary)

A log return is the continuously compounded return

r(t) as continuous return — compound n times a year and let n → ∞

Log return ⇔ continuous

$$r_t = \log(p_t) - \log(p_{t-1})$$

$$\Rightarrow \quad e^{r_t} = \frac{p_t}{p_{t-1}}$$

$$\Rightarrow \quad p_t = e^{r_t}\, p_{t-1} = \lim_{n\to\infty} \left(1 + \frac{r_t}{n}\right)^n p_{t-1}$$

How to read it

- Exponentiate to get the price ratio
- Limit as compounding count n → ∞
- Hence 'continuous return'
- The standard unit in finance

In ARCH · GARCH, r(t) means this log return

01

Why model volatility at all

Why model volatility? — big moves arrive in clusters

From the 1980s, econometrics shifted from the mean to volatility

History lesson — econometrics = economics using statistics

~1980s

Era of the mean

- Forecast the series' value itself
- AR · MA · ARMA · VAR

1982 · 1986

Volatility arrives

- Engle: ARCH (2003 Nobel Prize)
- Bollerslev: GARCH

NOW

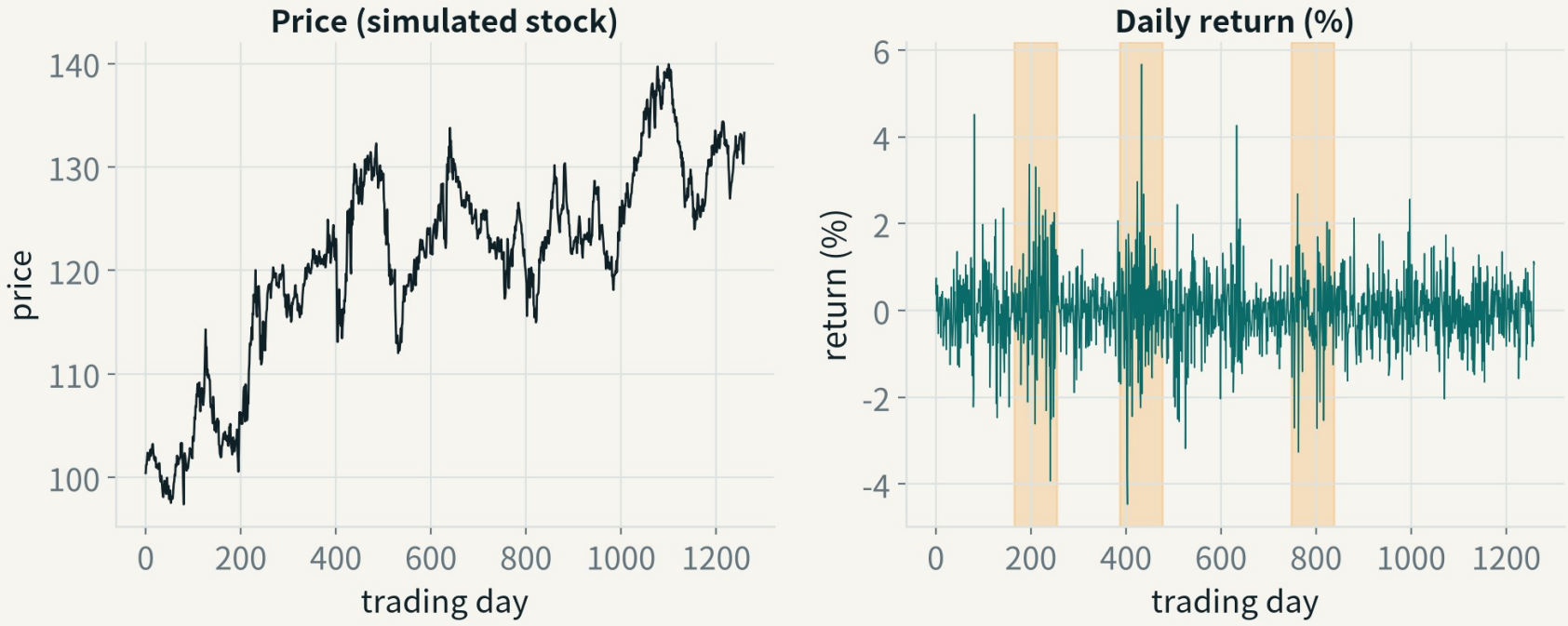
Volatility-centred

- What drives volatility?
- Volatility's effect on the mean

Risk, option pricing and allocation all start from volatility forecasts

Returns alternate between calm and turbulent spells

Why model volatility? — 5 years of simulated prices (GARCH); yellow bands: clustered big moves



Volatility clustering: big changes follow big changes, small follow small

A nuisance in cross-sections, a target in time series

Heteroskedasticity = error variance differs across observations

Cross-section vs series

$$\underbrace{\mathrm{Var}(\varepsilon_i \mid \mathbf{x}_i) = \sigma_i^2}_{\text{cross-section: nuisance}} \qquad \underbrace{\mathrm{Var}(\varepsilon_t \mid \mathbf{I}_t) = \sigma_t^2}_{\text{time series: of interest}}$$

How to read it

- Cross-section: correct it away
- Time series: forecast σ(t)
- I(t): information up to t
- Central in finance

The 'H' in ARCH is this Heteroskedasticity

Unconditional variance is one number; conditional variance moves with information

Conditional vs. unconditional variance

Unconditional: the intro-statistics variance

$$\mathrm{Var}(x) = \sigma^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 = E\Big(x - E(x)\Big)^2$$

Conditional variance: uncertainty given information I

$$\mathrm{Var}(x \mid \mathbf{I}) = \sigma^2_{\mathrm{cond}} = E\Big(x - E(x \mid \mathbf{I})\Big)^2$$

How to read it

- Unconditional: average swing overall
- Conditional: reflects what we know now
- Heteroskedastic: variance depends on factors
- ARCH · GARCH are conditional variance models

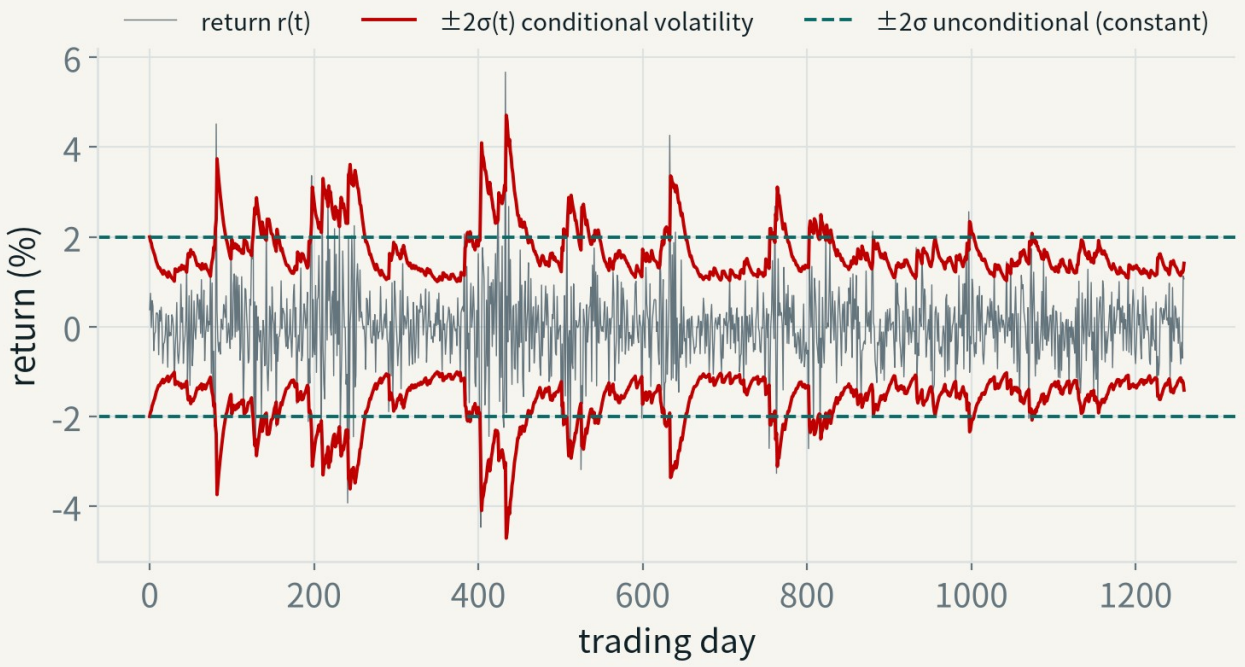
The question is not 'how much does it usually swing' but 'how much tomorrow'

Conditional volatility widens and narrows with clusters

Same simulation — red: ±2σ(t) conditional; green dotted: ±2σ unconditional (constant 1%)

How to read it

- Constant band: too narrow in crises
- Too wide in calm periods
- σ(t) must move over time



ARCH and GARCH are the equations that make σ(t) move

02

The ARCH Model

AutoRegressive Conditional Heteroskedasticity

ARCH forecasts tomorrow's variance from today's squared return

ARCH = time-varying volatility (conditional heteroskedasticity), modelled autoregressively

ARCH(1)

$$\underbrace{\sigma^2_{t+1}}_{\text{volatility tomorrow}} = \alpha_0 + \alpha_1 \underbrace{r^2_t}_{\text{“actual” today}}$$

How to read it

- Left: tomorrow's variance
- Right: today's squared return
- α0, α1: coefficients to estimate
- Big shock → tomorrow's variance ↑

AutoRegressive: variance regresses on its own past (squared returns)

A day's 'realised' variance is that day's squared return

Volatility of a single day? — in the variance formula, mean daily return ≈ 0, and with one observation (t = 1)

Why squared return = daily variance

$$\sigma_t^2 = \frac{1}{t}\sum_{i=1}^{t}(r_i - \bar{r})^2 \overset{\bar{r}\approx 0}{=} \frac{1}{t}\sum_{i=1}^{t} r_i^2 \overset{t=1}{=} r_t^2$$

How to read it

- Mean return ≈ 0 → the subtracted term vanishes
- One observation → one square remains
- So ARCH's right side is the squared return

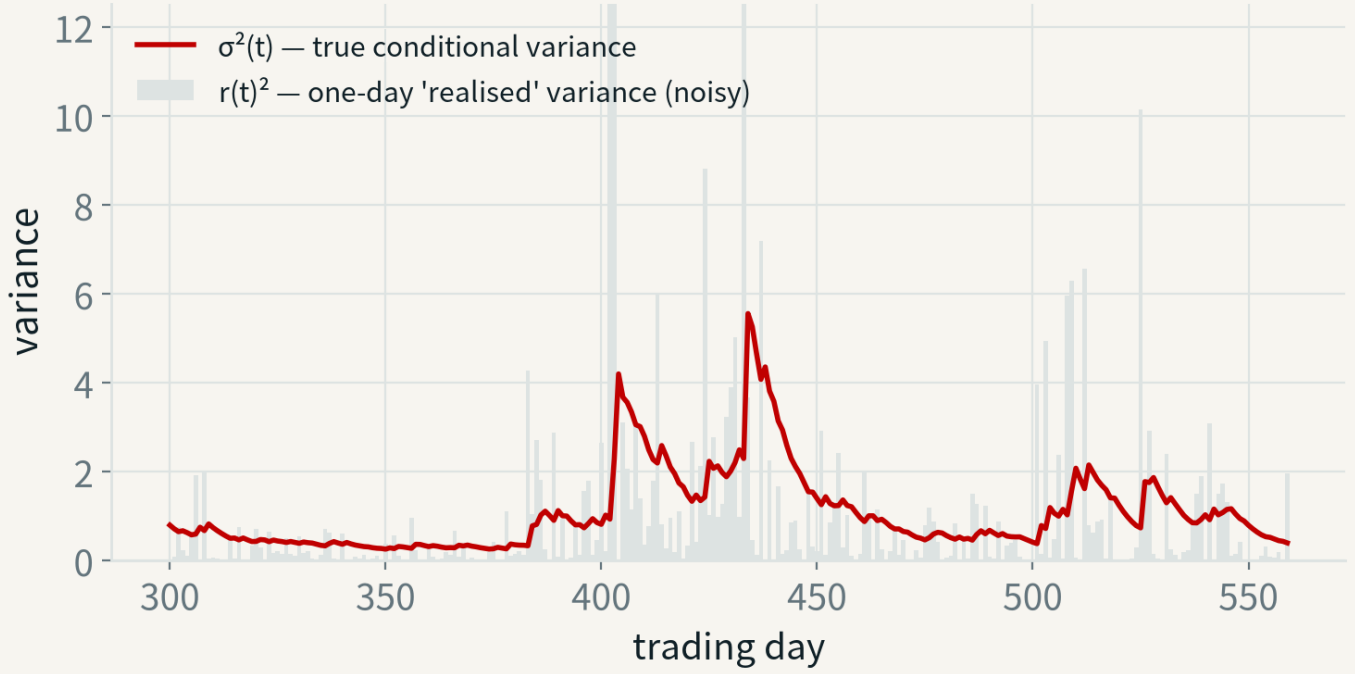
Squared return is 'observed' variance; the left side is 'forecast'

Squared returns are noisy but track the true variance on average

Part of the same simulation — grey bars: daily squared returns; red: true conditional variance

How to read it

- Daily values are jagged
- Variance is still high where big values cluster
- So α sets how much to trust them



ARCH estimates true variance from noisy observations (squared returns)

ARCH(q) uses squared returns from several past days

Can use multiple lags in the past — today, yesterday, … (q−1) days ago

ARCH(q)

$$\sigma^2_{t+1} = \alpha_0 + \underbrace{\alpha_1 r_t^2}_{\text{today}} + \underbrace{\alpha_2 r^2_{t-1}}_{\text{yesterday}} + \cdots + \underbrace{\alpha_q r^2_{t-q+1}}_{(q-1)\text{ days ago}}$$

How to read it

- Larger α: the day's shock lingers longer
- Real data need a large q
- → too many coefficients
- → GARCH solves it

Real world data needs LARGE q — the limit of ARCH

Coefficients must be positive, summing below 1, for stationary variance

Conditions and examples — numbers are daily returns in %

Conditions · long run · example

$$\alpha_0 > 0, \quad \alpha_i \geq 0, \quad \alpha_1 + \cdots + \alpha_q < 1 \;\Rightarrow\; \bar{\sigma}^2 = \frac{\alpha_0}{1 - \alpha_1 - \cdots - \alpha_q}$$

$$\text{example } (\alpha_0 = 0.1,\ \alpha_1 = 0.4):$$

$$r_t = 3\%: \quad \sigma^2_{t+1} = 0.1 + 0.4 \times 9 = 3.7 \;\Rightarrow\; \sigma_{t+1} \approx 1.9\%$$

$$r_t = 0.5\%: \quad \sigma^2_{t+1} = 0.1 + 0.4 \times 0.25 = 0.2 \;\Rightarrow\; \sigma_{t+1} \approx 0.45\%$$

How to read it

- Variance ≥ 0
- Sum ≥ 1: it explodes
- Today 3% → tomorrow σ 1.9%
- Today 0.5% → 0.45%

Forecast volatility rises after a big move = clustering reproduced

03

The GARCH Model

Generalised ARCH — widening ARCH as AR widened to ARMA

GARCH adds one term: yesterday's forecast variance

Similar to AR model extending to ARMA — uses actual observations (squared returns) and past forecasts (variance)

GARCH(1,1)

$$\underbrace{\sigma_{t+1}^2}_{\text{volatility tomorrow}} = \alpha_0 + \alpha_1 \underbrace{r_t^2}_{\text{“actual” today}} + \beta_1 \underbrace{\hat{\sigma}_t^2}_{\text{forecasted today}}$$

How to read it

- α1: weight on today's shock
- β1: inertia of the past forecast
- One of each usually suffices in real data
- → GARCH(1,1)

Real world data 'typically' only needs one of each

Unrolling yesterday's forecast turns GARCH(1,1) into an infinite ARCH

Substitute for $\hat{\sigma}^2(t)$ repeatedly — the distant past keeps small non-zero weight

GARCH(1,1) = ARCH(∞)

$$\begin{aligned}\sigma_{t+1}^2 &= \alpha_0 + \alpha_1 r_t^2 + \beta_1 \left(\alpha_0 + \alpha_1 r_{t-1}^2 + \beta_1 \hat{\sigma}_{t-1}^2\right) \\ &= \frac{\alpha_0}{1-\beta_1} + \alpha_1 \left(r_t^2 + \beta_1 r_{t-1}^2 + \beta_1^2 r_{t-2}^2 + \cdots\right)\end{aligned}$$

How to read it

- Weights α1, α1β1, α1β1², …
- Fading with age
- 3 coefficients, long memory

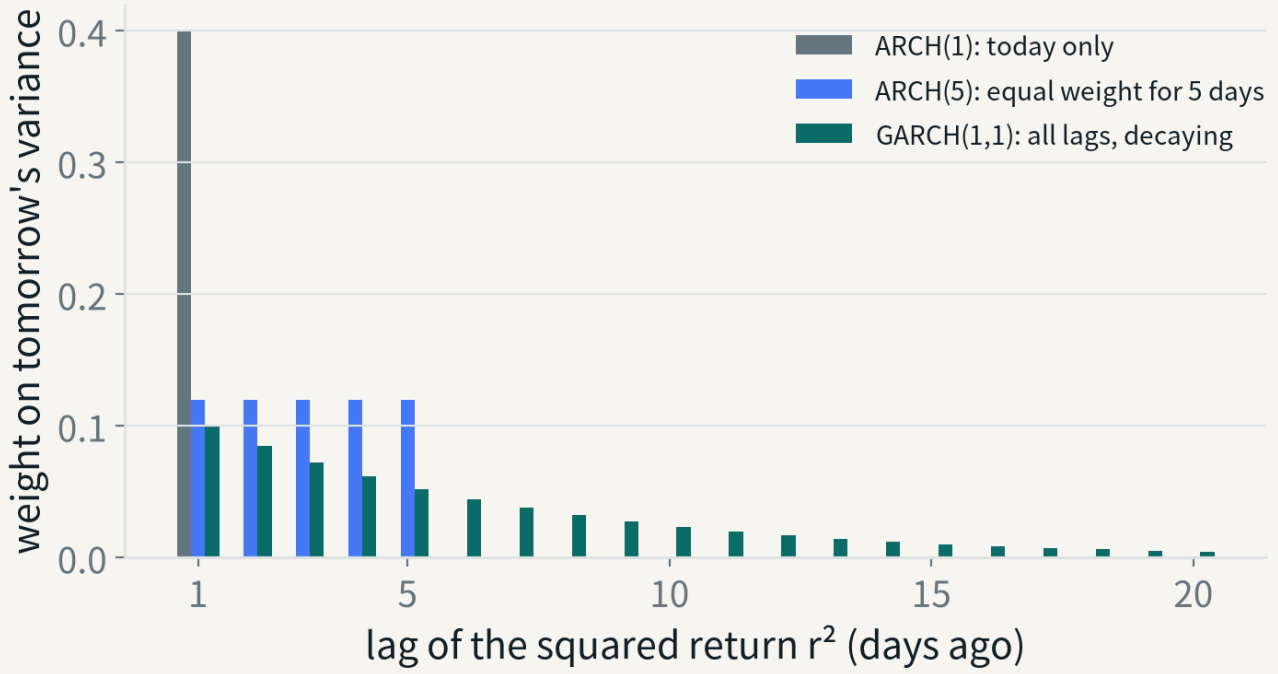
What ARCH does with q coefficients, GARCH does with 3

ARCH stops at a fixed day; GARCH weighs the whole past, fading

Weights of past squared returns on tomorrow's variance ($\alpha 1 = 0.10$, $\beta 1 = 0.85$)

How to read it

- ARCH(1): today only
- ARCH(5): cut off after 5 days
- GARCH: decays smoothly



Choose weights with 'ARMA-like' approaches — the heart of GARCH

α1 + β1 sets shock persistence; long-run variance sets where it returns

Persistence and mean reversion — for daily equity data, α1 + β1 is usually 0.95–0.99

Persistence · long run · half-life

$$\text{persistence: } \alpha_1 + \beta_1 < 1, \qquad \bar{\sigma}^2 = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}, \qquad h_{1/2} = \frac{\ln 0.5}{\ln(\alpha_1 + \beta_1)}$$

$$\text{example } (\alpha_0 = 0.05,\ \alpha_1 = 0.10,\ \beta_1 = 0.85):$$

$$\alpha_1 + \beta_1 = 0.95, \quad \bar{\sigma}^2 = \frac{0.05}{0.05} = 1\ (1\%/\text{day}), \quad h_{1/2} = \frac{\ln 0.5}{\ln 0.95} \approx 13.5 \text{ days}$$

How to read it

- Closer to 1: lasts longer
- Long-run σ 1%/day ≈ 16%/year
- Half a shock fades in about 2 weeks

Volatility clusters but returns to its long-run level

GARCH forecasts start at current volatility and converge to long-run variance

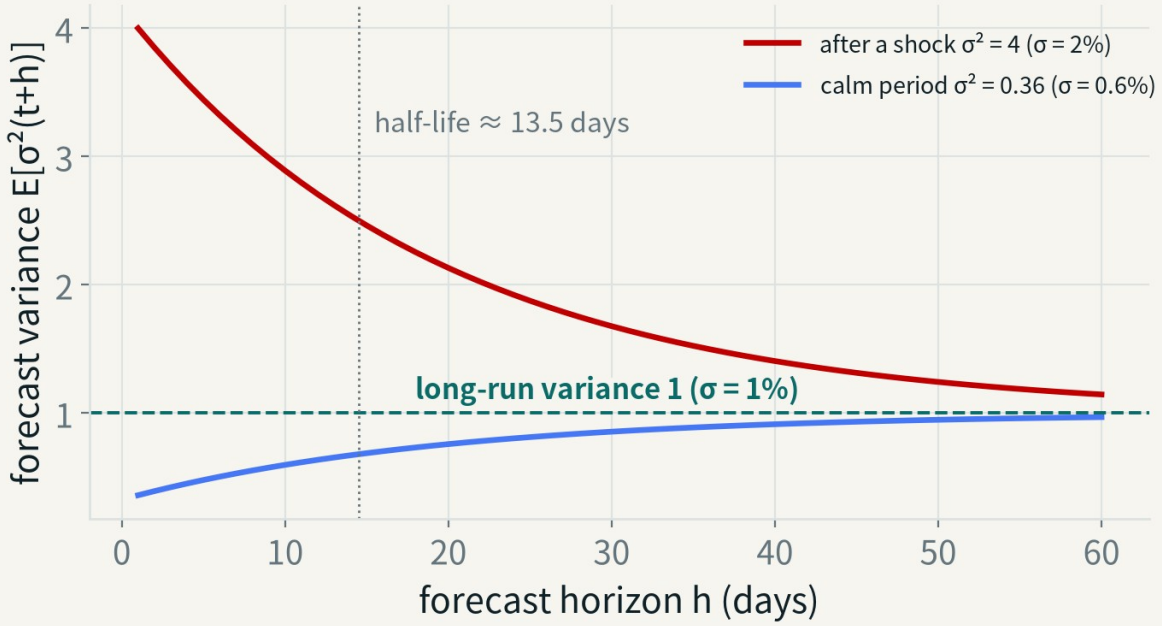
Same example ($\alpha 1 + \beta 1 = 0.95$, long-run variance 1) — forecasts after a shock and in calm

h-day-ahead variance

$$E_t\left[\sigma^2_{t+h}\right] = \bar{\sigma}^2 + (\alpha_1 + \beta_1)^{h-1}\left(\sigma^2_{t+1} - \bar{\sigma}^2\right)$$

How to read it

- Starting high: drifts down
- Starting low: drifts up



Short-term forecasts sit near 'now'; long-term near the 'mean'

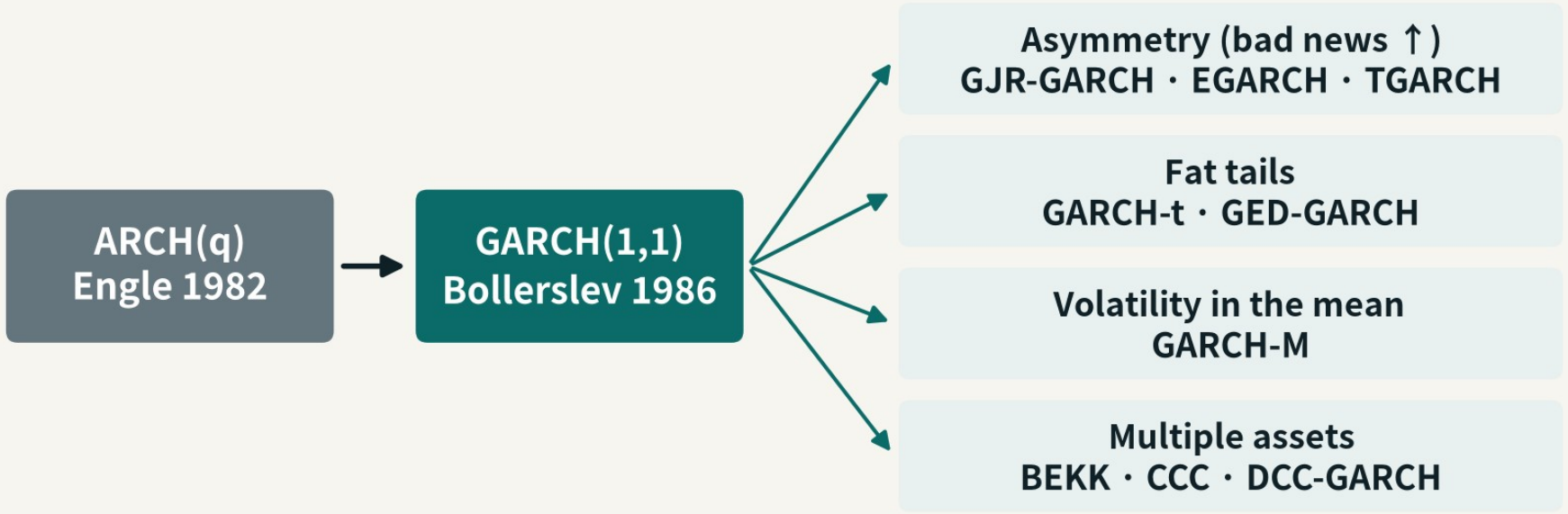
04

In practice

Many, many GARCH models — and estimating and using them

There are over a hundred GARCH variants; practice uses only a few

Many, many GARCH models — AARCH to ZARCH; four branches by purpose



Start with GARCH(1,1); extend to asymmetry, fat tails or multivariate when needed

In equities, bad news raises volatility more — an asymmetry

Leverage effect — a fall raises volatility more than an equal rise

GJR-GARCH: adds γ for bad news (negative returns)

$$\sigma^2_{t+1} = \alpha_0 + (\alpha_1 + \gamma\, I_t)\, r_t^2 + \beta_1\, \hat{\sigma}_t^2$$
$$I_t = 1 \text{ if } r_t < 0, \quad 0 \text{ otherwise}$$

How to read it

- If r(t) < 0, weight is α1 + γ
- γ > 0 confirms asymmetry
- EGARCH serves the same aim (log variance)
- Almost always significant for indices

Explaining volatility spikes in crashes needs an asymmetry term

Coefficients are estimated by maximum likelihood — one line of code

MLE, not OLS — assume returns are normal with mean 0 and the conditional variance, and maximise the likelihood

Normal log-likelihood

$$\ln L = -\frac{1}{2}\sum_t \left(\ln 2\pi + \ln \sigma_t^2 + \frac{r_t^2}{\sigma_t^2}\right)$$

How to read it

- Python: the arch package
- arch_model(r, vol='GARCH', p=1, q=1)
- Output: α0, α1, β1 and σ(t)
- Fat tails: dist='t'

Enter returns in % — decimals make estimation unstable

Forecast σ(t+1) feeds straight into VaR and volatility-target weights

In practice — input to risk limits, position sizing and option prices

Volatility-target weight and 1-day 99% VaR

$$w_t = \frac{\sigma_{\text{target}}}{\sigma_{t+1}}$$

$$\text{VaR}_{99\%,\,t+1} = 2.33\,\sigma_{t+1} \times \text{position}$$

How to read it

- σ(t+1) 2%, target 1% → weight 50%
- Cut when volatility is high, add when low
- VaR: 2.33σ under normality

Better volatility forecasts give steadier performance for the same risk

05

Summary

Stack a variance model on a mean model

ARCH uses shocks; GARCH uses shocks and memory

Mean models (AR·ARMA) vs variance models (ARCH·GARCH)

	Mean model	Variance model
Forecast target	Value of the return	Conditional variance
Past observations	AR: past values	ARCH: past squared returns
Observed + forecasts	**ARMA: past values + past errors**	**GARCH: past squares + past variance**
Workhorse in practice	AR(1) · ARMA(1,1)	GARCH(1,1)

Next week: VAR, many series together

Stationary returns, clustered volatility

ARCH — yesterday's shock, today's volatility

GARCH — long memory from 3 coefficients

Next week
VAR models